

Windows Exploit | Exploitation of Remote WCF Vulnerabilities

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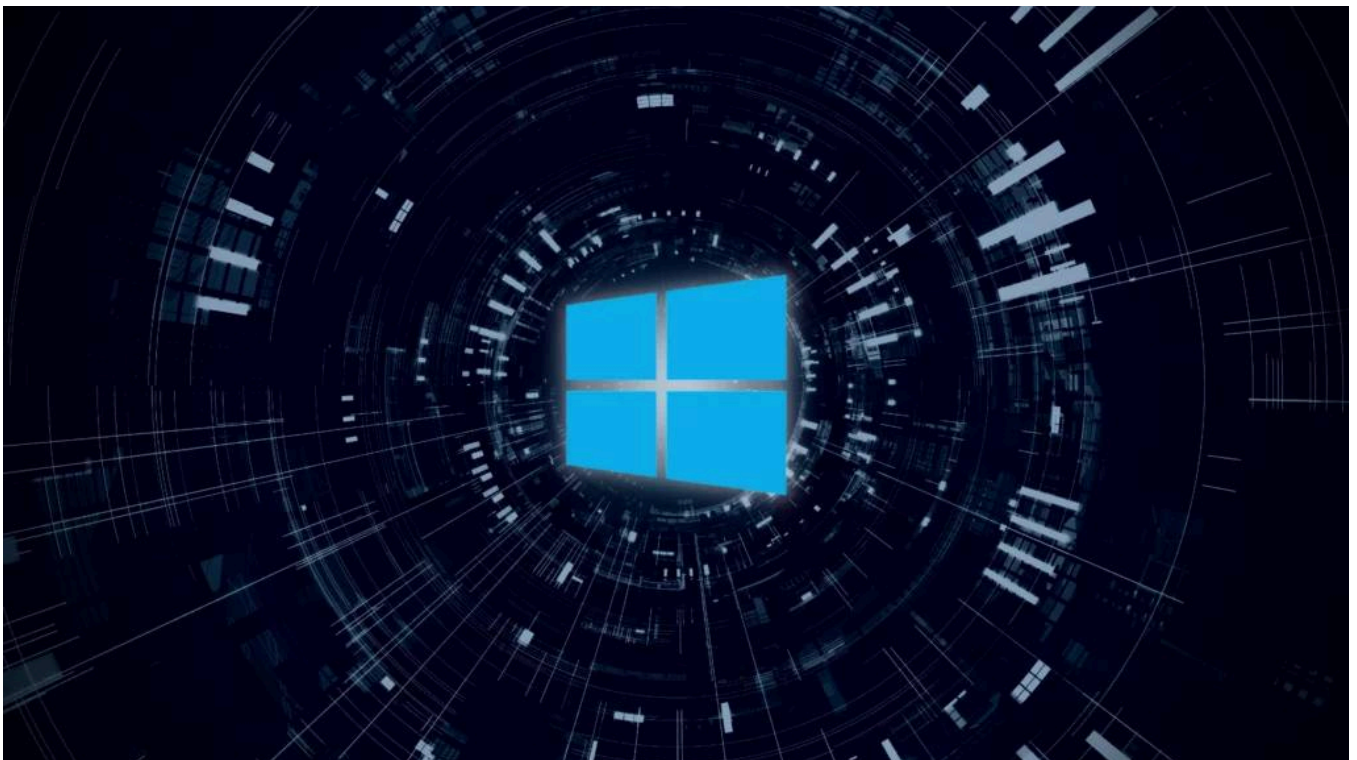
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Content

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Exploitation of Remote WCF Vulnerabilities



What is WCF Services? Remote WCF Exploitation: CVE-2019-8917

[Video](#)

Remote WCF Exploitation: CVE-2019-8917

In 2018, VerSprite's research team invested a significant amount of time into **abusing insecure Windows Communication Foundation (WCF) endpoints**.

In our previous blog, we discuss the analysis and exploitation of this vulnerability class. In addition, we presented **our research at Ekoparty 2018** where we alluded to the exploitation of a remotely accessible WCF endpoint vulnerability. Until now, we've not discussed the exploitation of remote WCF vulnerabilities in detail.

In this blog, we'll be discussing the discovery, analysis, and exploitation of **CVE-2019-8917** which is a remotely exploitable WCF vulnerability that we discovered in **SolarWinds Orion NPM 12.3**.

This issue was patched in the 12.4 release of SolarWinds Orion NPM which was made available in December of 2018.

Discovery: Remote Exploitable WCF

We were inspired to discover a remotely exploitable WCF vulnerability that could be shared as an example of the remote attack surface introduced by insecure endpoints. We searched Google for results containing the error message **"there is no endpoint listening at net.tcp"** as is defined by .NET's **EndpointNotFound** error.

Our goal was to identify applications known to leverage WCF's **NetTcpBinding** to expose a remote endpoint. This query led us to discover that SolarWinds exposes a remote WCF endpoint on port 17777 within all versions of NPM.

SolarWinds Orion NPM (**Network Performance Monitor**) is a software designed to reduce network outages and improve performance. As we sought to learn more about the WCF service exposed by NPM, we downloaded and installed the free trial edition within a virtual machine. We then reviewed the details of the service named **OrionModuleEngine** which was installed alongside NPM:

Note that this service uses **"NetTcpPortSharing"**.

This service is also started as **"LocalSystem"** which suggests that exploiting any of the WCF services that may be exposed by ****OrionModuleEngine**** will provide us with SYSTEM privileges.

This is a component of WCF that allows multiple services from different processes to share a single "net.tcp" port. Using **dnSpy**, we decompiled the assembly found at **"C:\Program Files (x86)\SolarWinds\Orion\SolarWinds.BusinessLayerHost.exe"** for analysis.

We found that when the service is started, it first handles license verification, then attempts to detect and load "plugins" immediately after. **Log4net** is used throughout this service to log the entire process to

"C:\ProgramData\SolarWinds\Logs\OrionBusinessLayerHost.log". Reviewing this log file revealed to us the plugins that had been loaded:

```
...

SolarWinds.BusinessLayerHost.PluginManager - Discovered plugins:

**SolarWinds.Orion.Core.BusinessLayer.dll**

SolarWinds.Orion.ServiceDirectory.BusinessLayer.dll

Solarwinds.Settings.BusinessLayer.dll

SolarWinds.AddNode.BusinessLayer.dll

SolarWinds.AgentManagement.ServiceCore.dll

SolarWinds.AgentManagement.Watchdog.dll

SolarWinds.ASA.BusinessLayer.dll

SolarWinds.Orion.Cli.BusinessLayer.dll

...
```

According to the decompiled source code, when the plugin associated with **“SolarWinds.Orion.Core.BusinessLayer.dll”** is loaded, a new instance of **CoreBusinessLayerService** is started.

The service definition found in **“SolarWinds.Orion.Core.BusinessLayer.dll.config”** informs us that there are several endpoint addresses associated with this service, including the following remotely accessible endpoint:
“net.tcp://localhost:17777/orion/core/businesslayer”

The associated Service Contract we reviewed, **ICoreBusinessLayer**, exposed 383 different methods. Following a cursory analysis of several methods, we decided to focus our attention on the method named **InvokeActionMethod**.

Analysis

The server-side implementation of **InvokeActionMethod** is defined in the **ActionMethodInvoker** class of the **SolarWinds.Orion.Core.Actions** namespace.

InvokeActionMethod attempts to match the user provided **actionTypeID** string against **actionTypeID** values associated with a known **ActionMethodInvoker**. If a matching **ActionMethodInvoker** is found, then it's **InvokeMethod** method is called using the user provided **methodName**, and **args** arguments. Reviewing the decompiled source code for **ActionTypeID** strings revealed the following.

- CustomProperty
- CustomStatus
- Email
- EmailUrl
- EmailWebPage
- ExecuteExternalProgram
- ExecuteVBScript
- FailoverAction
- Geolocation
- NetMessage
- NotePage
- PlaySound
- Print
- SaveToDisk
- SendHttpRequest
- SendSyslog
- SnmpTrap
- TextToSpeech
- WriteToEventLog
- WriteToFile
- WriteToNPMEventLog

Each of the **ActionTypeID** strings correspond to an implementation namespace within **SolarWinds.Orion.Core.Actions**. Some of these namespaces include an invoker class that implements **IActionMethodInvoker**, which is required by **InvokeActionMethod**. This leaves us with the following action namespaces:

- Orion.Core.Actions.Impl.Email
- **Orion.Core.Actions.Impl.ExecuteExternalProgram**
- **Orion.Core.Actions.Impl.ExecuteVBScript**
- Orion.Core.Actions.Impl.NetMessage
- Orion.Core.Actions.Impl.NotePage
- Orion.Core.Actions.Impl.PrintURL
- Orion.Core.Actions.Impl.SaveToDiskURL
- Orion.Core.Actions.Impl.SnmpTrap
- Orion.Core.Actions.Impl.WriteToFile

Obviously, the actions whose names begin with “Execute” immediately caught our attention. Next, we reviewed the decompiled source of the **ExecuteExternalProgramInvoker** class to determine which values should be passed for **methodName** and **args**.

Based on the source of the invoker's **InvokeMethod** method, only one **methodName** is acceptable: **ValidateAccess**.

In addition, an instance of this action's configuration class, **ExecuteExternalProgramConfiguration**, is created from a deserialization of the **args** parameter. The Data Contract associated with this class defines Data Members for the program path and active directory credentials.

Initially, we feared that this was the end of the road for this approach, as we presumed that the active directory credentials would be used for access validation. However, if the **Credentials** member of the configuration object are left null, the **ValidateAccessToFile** method will simply pass the **ProgramPath** string to a method named **ExecuteWithoutCredentials**.

This method will then call **Microsoft.VisualBasic.Interaction.Shell** and proceed to execute our program as the current user, **"NT AUTHORITYSYSTEM"**.

Armed with knowledge of the **actionTypeID**, **methodName**, and **args** values required to reach an exploitable code path from the **InvokeActionMethod** method, our next step is exploitation.

Exploitation

Most of our WCF exploits have abused the **NetNamedPipeBinding**, gaining us privilege escalation. However, the **NetTcpBinding** was less familiar to us, so we consulted the decompiled source code for guidance on communicating with the service endpoint.

Although, we have remotely exploited insecure WCF endpoints in the past, we had yet to encounter any transport security, until now. This endpoint requires credentials in the form of a **UserName** and **Password** string in order to successfully establish a connection:

Luckily, the **GetPassword** method simply returns a Base64 encoded hash of the username that we provide. By leveraging the **MessageUtilities** class in our exploit, authentication becomes trivial.

Another roadblock we encounter was that attempting to connect to the target using an IP address resulted in the occurrence of a DNS identity error which terminated our connection.

Identity check failed for outgoing message. The expected DNS identity of the remote endpoint was '192.168.1.243' but the remote endpoint provided DNS claim 'MSEDGEWIN10'. If this is a legitimate remote endpoint, you can fix the problem by explicitly specifying DNS identity 'MSEDGEWIN10' as the Identity property of EndpointAddress when creating channel proxy.

As a workaround, we first attempted a connection using the target's IP address, then we parsed to DNS identity error message in or to retrieve the expected DNS identify. Finally, we attempted a second connection providing this new explicit endpoint identity as part of our **EndpointAddress**.

After this was sorted, we were able to successfully create a channel to the **CoreBusinessLayer** service. Next, we want to call the remote **InvokeActionMethod** method, but first we must populate the configuration object to be serialized as our **args** argument.

The command we chose for the **ProgramPath** Data Member was a PowerShell reverse connect one-liner from [Nishang](#), an open-source PowerShell framework for red teaming. A TCP listener is launched at the beginning of the exploit in preparation for receiving this reverse connect shell.

For the **Credentials** Data Member, we simply set the value to null to circumvent access validation, as explained earlier.

After our configuration object is populated, we call **InvokeActionMethod** with "ExecuteExternalProgram" as the **actionTypeID** argument, "ValidateAccess" as the **methodName** argument, and we serialize our configuration object to an XML string so that it may be used as our **args** argument.

Moments after our exploit is launched, we are greeted by a SYSTEM shell courtesy of the vulnerable target.

Remediation

When we reported this security issue to SolarWinds, their product security and incident response team were very receptive. We worked together to ensure that a patch would be made available prior to disclosure. As mentioned earlier, this issue is patched in the SolarWinds Orion NPM 12.4 release. In the patched version of this software, it is no longer trivial for arbitrary clients to authenticate to the service and call remote methods.

Timeline:

- **10-01-2018**– Disclosed to SolarWinds
- **10-02-2018**– Response from SolarWinds
- **10-08-2018**– Coordination of patch and disclosure with SolarWinds
- **12-04-2018**– Verified patch in 12.4 release

Remote WCF Endpoint Exploitation Conclusions

This case study of remote WCF endpoint exploitation should help to clarify the threat impact of this vulnerability class.

As the adoption of WCF continues to increase, so does the chance of exposure for insecure endpoints. We encourage you to help us spread awareness of this vulnerability class to security professionals and software developers alike.

If you'd like to read more on vulnerabilities and exploit development from our research team, be sure to check out our latest presentation on [Abusing Insecure WCF Endpoints](#).

Security Research

VerSprite’s Research and Development division (a.k.a VS-Labs) is comprised of individuals who are passionate about diving into the internals of various technologies. From advanced technical security training to our research for hire B.O.S.S offering, we help organizations solve their most complex technical challenges.

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Identify Vulnerable WCF Services

Learn useful techniques to identify vulnerable WCF services, discover what to look for when analyzing decomposed .NET assemblies, including those that have been obfuscated, and watch a demonstration of attacks against real software.

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